

MAVE: Maze-based immersive virtual environment for new presence and experience

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Abstract

To provide users with a sense of presence and experience, this study presents a maze-based immersive virtual environment (MAVE). MAVE consists of a new immersive virtual scene and immersive interaction optimized for such maze. The algorithm for calculating a finite maze is defined to automatically generate diverse maze patterns; an interactive editing function and sketch-based maze retrieval make it easier to generate maze patterns intuitively. With the integration of these procedures, MAVE proposes a user-oriented maze terrain authoring system that can reconstruct two-dimensional maze patterns into three-dimensional maze terrains. It also provides an interaction system that enables users to be immersed in a broad virtual maze environment within a limited space. As an Arduino-based portable walking simulator, this system detects the user's steps as he or she walks in place and controls the user's motions in the virtual environment. This study confirms through various technical and statistical experiments that MAVE can lead to new research on immersion enhancement without virtual reality sickness via virtual reality content.

KEYWORDS

immersion, immersive virtual reality, interaction, maze generation algorithm, portable walking simulator, VR sickness

1 | INTRODUCTION

Virtual reality (VR) provides users with an experience similar to actual reality as it interacts with users through their senses within a virtual environment. By tracking the user's head and other motions, a VR system shows an appropriate display, and devices like three-dimensional (3D) displays and head-mounted displays (HMDs) are used to enhance a presence. Diverse sensors and devices, ranging from motion capture to treadmill systems, are being developed to accurately measure joints, muscles, gestures, and walking to capture user movement. Additionally, there have been various studies on haptic systems that deliver physical responses from the virtual environment. Such VR techniques have expanded users' feelings of extensity and interactivity, and immersion.

VR has been advanced by several studies. These include HMD system that transmits the 3D visual information of a VR scene,¹ an auditory/haptic system that maximizes the immersion by incorporating all five senses of the user, including auditory and tactile, and analysis of presence in terms of perception and consciousness.^{2–4} Haptic systems, in particular, are being developed from hand-centered tactile response using gloves to a system that provides feedback on walking.⁵ There has also been research on tracking user motion to send virtual scenes of the correct location and expressing user actions—physically touching or grabbing surrounding props—with enhanced user immersion.⁶ Likewise, there has been diverse research with the purpose of reconstructing what users see and feel to resemble actual reality.

One crucial factor to consider when creating an immersive virtual environment and providing users with a sense of

presence and experience is the virtual scene. However, research on hardware-centered VR, like haptic devices, has limited space in which to develop a scene, and the composition of scenes is also simple in many cases. Nevertheless, research on creating complex virtual spaces is essential because VR technology is applied to many areas, ranging from fine arts to architectural design. To this end, there are studies on the automatic creation of unique virtual scenes that are complex in structure, including urban spaces or virtual landscapes in procedural structures.^{7,8}

Therefore, this study proposes a method of creating a virtual maze scene through an immersive virtual environment by providing users with a sense of presence and experience. This study also proposes a new simulator that can enable immersive interaction within the proposed virtual maze space.

Maze-based immersive virtual environment (MAVE) makes the following contributions:

- A new integrated authoring system that is equipped with a maze generation algorithm, interactive editing, and sketch-based maze retrieval to easily and automatically create maze terrains.
- A novel portable walking simulator is proposed, so that the system can express immersive interaction as the user walks within the limited maze space.

Through technical and statistical experiments, MAVE analyzes and verifies system satisfaction and VR/simulator sickness.

Section 2 begins with previous VR research, including virtual scenes and haptic systems. Section 3 explains the maze terrain authoring system and immersive interaction MAVE system. Through various tests, in Section 4, the effects of immersive interaction and the process of creating a VR maze terrain are analyzed through the effect MAVE has on users. Then, Section 5 discusses the limitations of this research. Conclusion and future research directions are explained in Section 6.

2 | PREVIOUS WORK

Aiming to stimulate users' senses to provide spatial and temporal experiences similar to those of actual reality, there has been expansive research on VR. Sutherland¹ studied a head-mounted VR system, a device that sends virtual visual information to the user dimensionally and realistically. Research on VR later evolved to study haptic systems; motion platforms; and VR simulators that enhance user immersion, realism and presence.^{2,9,10}

Research related to haptic interfaces has also been conducted to enable users to interact with virtual environments using their hands and to receive tactile sensations through force feedback interfaces, like electrical actuators.¹¹

Carvalho et al² studied a haptic interaction system that provides users with the experience of controlling an object in a realistic virtual environment by combining a real-to-virtual mapping system and a device that tracks the hand and physical object movement. Additionally, researchers have also studied ways to enhance immersion within virtual environments by providing tactile feedback to the user, as with low-cost VR-based smart gloves.¹² Such haptic systems have also been applied to other body parts. Nordahl et al⁵ designed a system that provides users with the experience of walking on actual ground in virtual environments through haptic shoes and sound synthesis. Suggesting a vibrotactile device that displays the virtual ground material, Visell et al¹³ expressed the feeling of walking on the ground realistically. This, however, is not about tracking the users' motions but only provides a sense of what it is like to walk by providing tactile feedback of walking on the ground.

Motion platforms have been applied to enhance the realism of virtual interaction by capturing user's expressions and motion. The Haptic Turk system¹⁰ provides experiences like flight simulation by receiving help from a human actuator. There are also applicative studies that have expanded the scope of user movement and gesture within a limited space. Cheng et al⁶ suggested a prop-based VR system in which a user can not only see and listen but also feel. By providing an experience that is similar to actually controlling diverse props, including doors and walls, within a virtual space, this system provides users with an enhanced sense of immersion. The users' movement, however, is limited within virtual environments. Vasylevska et al¹⁴ suggested a flexible space method using dynamic layout generation as a means for infinite walking within a virtual environment. However, there are still limitations and constraints to such work. Therefore, future research that can help resolve VR spatial limitations is needed.

To provide users with new senses of presence and experience, satisfying them visually through virtual scenes is as important as accurately detecting user movement and providing feedback. Studies have been conducted to demonstrate that visual realism has a crucial impact on presence in a VR environment.¹⁵ Going forward, VR scenes must move beyond indoor and other limited spaces and be able to express urban or other outdoor landscapes magnificently. Vanegas et al⁷ conducted various surveys on the realistic expression, effective modeling, and visualization of urban spaces that consist of complex collections of buildings, parcels, blocks, and streets. This study compared and analyzed various approaches, including procedural methods and image-based modeling.^{16,17} Dylla et al¹⁸ used procedural modeling to study the reconstruction of Rome as a virtual city. Kim⁸ also studied ways of effectively expressing VR-based landscape by automatically generating diverse trees through procedural modeling. Most such research, however, considers real-time

systems, like games, as their application boundaries and does not account for the technical factors (refresh rate, fps, etc.) that are required from the VR system to prevent side-effects like VR sickness.

3 | MAZE-BASED IMMERSIVE VIRTUAL ENVIRONMENT

3.1 | System overview

This study proposes a maze-based immersive virtual environment, MAVE, to provide a new sense of presence and experience to users. MAVE can be divided into two types of technique: First, there is an authoring system that supports the creation of 3D maze virtual scenes. When users experience the same maze pattern repeatedly, their interest drops dramatically. Therefore, the core objective of the authoring system is to automatically generate diverse maze patterns or to easily devise a maze pattern desired by the user. To this end, the system has three processing procedures: a maze generation algorithm, interactive editing, and sketch-based maze retrieval. Second, there is immersive interaction in which users can directly experience escaping the maze. A portable walking simulator using gyro sensors is built for this process. The user wears the simulator on an ankle; when the user marches in place, he or she moves around in the virtual maze space. The core of the proposed interaction is to provide an immersive environment more fit for maze terrain compared to simple input controllers like keyboard, without using expensive and complex devices. Figure 1 shows an overview of the MAVE system.

Using the proposed authoring system to calculate the information necessary for creating maze terrain, graphic factors must be added to the virtual maze space as the simulator moves the character and the final rendered scene must be transmitted via HMD. To this end, this study creates a virtual environment by establishing a Unity3D engine-based integrated development environment using the maze information from the our authoring system, walking simulator, and Oculus software development kit (SDK).

3.2 | Maze terrain authoring system

The maze terrain authoring system in this study has three objectives. First, it automatically calculates mazes of diverse sizes and patterns. Second, it edits maze patterns as desired by the user. Lastly, it inputs hand-drawn mazes to the system to grant a variety of patterns.

The first function is automatically generating mazes of diverse size and pattern. One crucial element is the fact that the system needs to generate a finite maze where only one route exists. Acyclic mazes, where the start and end points are very clear, can provide a virtual maze environment for the users. Among the many maze generation algorithms, this research is unique in that it uses Jeong and Kim's maze algorithm,¹⁹ which generates a route centering on the starting point, and thus provides an experiential environment by corresponding the starting point of the newly created 3D virtual maze. This method applies the process of calculating the minimum spanning tree by gradually expanding the spanning tree set in the beginning route onto the maze pattern. The algorithm generates roads and walls through the recursive analysis of two-dimensional (2D) grid type cells. During this process, random numbers are applied to generate new maze patterns. The unique aspect of this method is that it does not generate walls that limit users from entering and circulating routes.

The proposed finite maze generation algorithm used herein is composed as follows. First, the user sets the size (width $\times$ height) of the 2D grid cells that compose the maze. Here, all cells are selected as either walls or roads; the cells selected as road cells comprise the maze route. The first procedure selects the starting cell (p_c^s), which can either be selected randomly from among the 2D cell sets or by the user. (a) The selected cell becomes the starting point of the road. Then, (b) one of the four neighboring cells of the starting cell is randomly selected, determining the road cell (p_c^r). When a cell is determined as the road, a four-direction test of the surrounding cells is conducted (**NEIGHBORING CELLS SEARCH**). (c) This test registers the neighboring four cells as a wall cell set, and the road cell becomes the parent of the wall cell. The cell already selected as a road cannot become a wall.

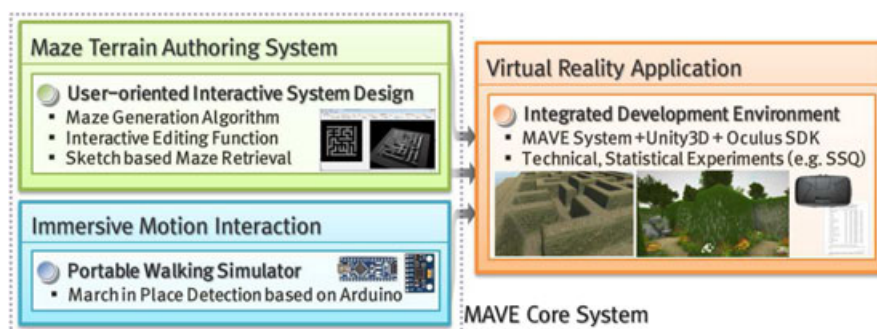


FIGURE 1 System overview of the maze-based immersive virtual environment (MAVE)

Then, (d) the next road is determined by taking down one (p_c^c) of a neighboring wall cell and the current road cell ($p_c^{pr}(=p_c^r)$) to calculate direction (v^r), and (e) from the current road and through the open wall is a connecting cell ($p_c^r = p_c^{pr} + v^r$) that will be defined. When the new road cell is determined, steps c–d are processed recursively to select road and wall cells. Algorithm 1 shows the finite maze automatic generation algorithm.

Algorithm 1 Finite maze auto-generation algorithm.

```

1:  $n[2] \leftarrow$  define the size of the 2D maze.
2: Array_walls  $\leftarrow$  a set of wall cells.
3: procedure FINITE MAZE GENERATION  $n$ , Array_walls
4:   (1)  $p_c^s$  : setting the start point of the maze ( $n$ ).
5:   (2)  $p_c^r \leftarrow$  road cell:  $p_c^s$  selecting one of the neighboring
      four cells.
6:   while Array_walls can be computed do
7:     (3) NEIGHBORING CELLS SEARCH ( $p_c^r$ , Array_walls)
8:     (4)  $p_c^c \leftarrow$  take one cell out of Array_walls.
9:     (4)  $\vec{v}_r = p_c^c - p_c^{pr}$  : compute the direction of the road.
10:    (5)  $p_c^r = p_c^c + \vec{v}_r$  : compute the next road.
11:   end while
12:   Set the cells of Array_walls as walls and the rest as
      roads.
13: end procedure
14: procedure NEIGHBORING CELLS SEARCH  $p$ ,
      Array_walls
15:   Array_Cwalls[4]  $\leftarrow$  set the four directions around  $p$ 
      as the cells of the candidate walls.
16:   for  $i=0, 3$  do
17:     if Array_Cwalls[ $i$ ] is road then
18:       exclude it from the Array_walls.
19:     else
20:       Array_Cwalls[ $i$ ] is stored in Array_walls.
21:       set  $p$  as the parent cell of Array_Cwalls[ $i$ ].
22:     end if
23:   end for
24: end procedure

```

Based on the defined algorithm, a 2D maze pattern is calculated, as seen in Figure 2a. This pattern must be reconstructed into a 3D maze terrain. This study uses heightmap-based terrain creation to propose a method that automatically generates a maze texture map, transforming the 2D cell information distinguished into wall and road cells into same-sized images (Figure 2b). A factor that must be considered here is that the automatically calculated maze pattern may not always be an enclosed wall (Figure 2c). In this case, setting the destination could be unclear; thus, the method generates enclosed walls in all conditions. The next destination point is transformed into a road, and the patterns are set to allow experience to the user (Figure 2c). When the calculated maze pattern is transformed into a maze image, a heightmap texture map that can

create 3D terrain is generated. The important aspect of this process is that the maze texture map to be used for the 3D terrain must satisfy 2^n resolution. However, when generating a texture map with the same ratio regardless of the pattern and size of the maze, the complicated maze pattern can be too narrow for users to go through (Figure 2d). Therefore, the texture map must be relative to the size of the maze pattern, always maintaining the same road and wall widths to provide a comfortable experience to users (Figure 2e). In this process, the border that can express the terrain surrounding the maze is also set. Equation 1 shows the process of transforming a user-defined maze pattern into a $2^n \times 2^n$ maze texture map coordinate.

$$\begin{aligned}
 I(x, y) &= C(x + (\kappa - w) + \alpha_{cw}^0 + \alpha_{br}, y + (\kappa - h) + \alpha_{cw}^1 + \alpha_{br}), \\
 T(x, y) &= I(\mu^0 * x + \alpha_{dx}, \mu^1 * y + \alpha_{dx}), \\
 \kappa &= w > h ? w : h, \\
 \mu^i &= \kappa / \kappa_s * v_s / (\kappa + \alpha_{cw}^i + 2\alpha_{br}), \\
 \text{where } i &= 0, 1.
 \end{aligned} \tag{1}$$

Here, α_{cw}^i refers to left/right or up/down enclosed wall additional value, α_{br} is border size, and κ represents maze pattern size. Moreover, κ_s is a maze pattern reference value and v_s is the reference value of the 2^n texture map. This study has set the maze pattern to 15 and texture to 512. μ^i is the relative size of the texture map, which considers maze patterns, and α_{dx} is the difference in expanding and transforming textures of relative sizes into a 2^n texture map. Based on this, the cell (C) of a 2D maze pattern is transformed into an image (I) while considering closed walls and borders. The image (I) is then transformed into a 2^n -sized maze texture map (T) that considers the relative size.

The second step of the proposed authoring system is the interactive editing function. Although the maze pattern can express the virtual maze environment, the user can edit the route to certain extent. Therefore, when a maze pattern image that is suitable for experience is generated through the 2^n maze texture map, users can interactively change the walls and roads. Further, the system is also equipped with an interactive function in which the user can set a starting point on the 3D maze virtual environment in advance. Equation 2 represents two processes. One process involves transforming the coordinates selected by the user on the authoring system window (W) into the corresponding coordinates on the maze pattern image (P), and then exchanging the wall and road cell. The other process involves mapping the click coordinates of the 2D maze pattern to the 3D maze coordinates (X) by considering the size of the maze and the resolution of the maze texture map.

$$\begin{aligned}
 P(x, y) &= W(x * (\kappa + \alpha_{cw}^0 / \omega^0), y * (\kappa + \alpha_{cw}^1 / \omega^1)), \\
 \text{State}(P) &= \text{State}(P) \text{ is wall ? road : wall}, \\
 X(x, z) &= T(x, y) * m / v.
 \end{aligned} \tag{2}$$

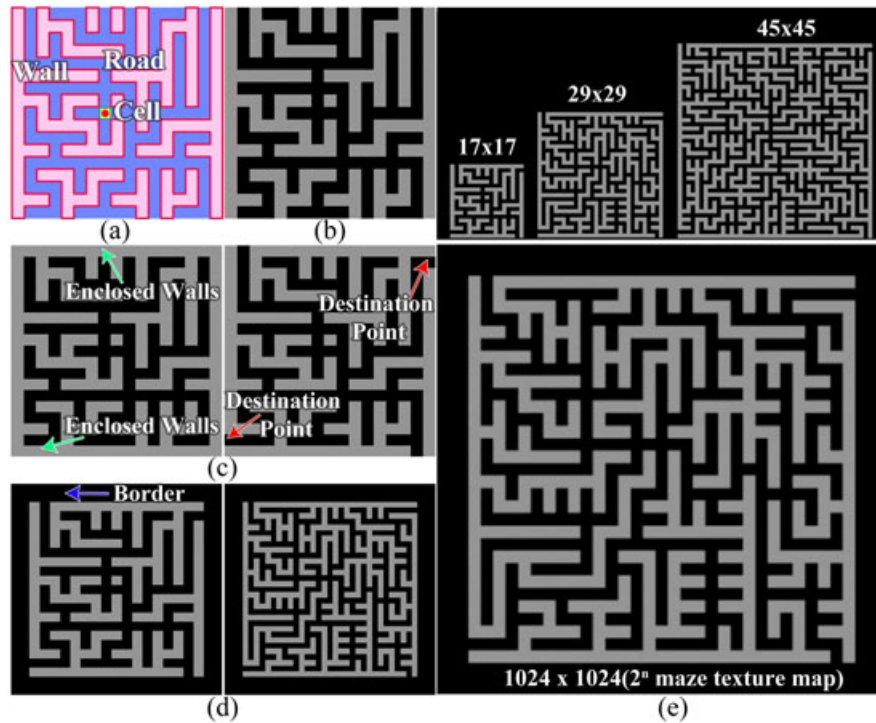


FIGURE 2 Generating maze texture maps based on a finite maze automatic generation algorithm: (a) maze pattern automatically calculated through classifying the walls and path, (b) results of maze pattern image transformation, (c) maze pattern image edited to enable user experience, (d) setting a relative texture map, and (e) completed 2^n maze texture map

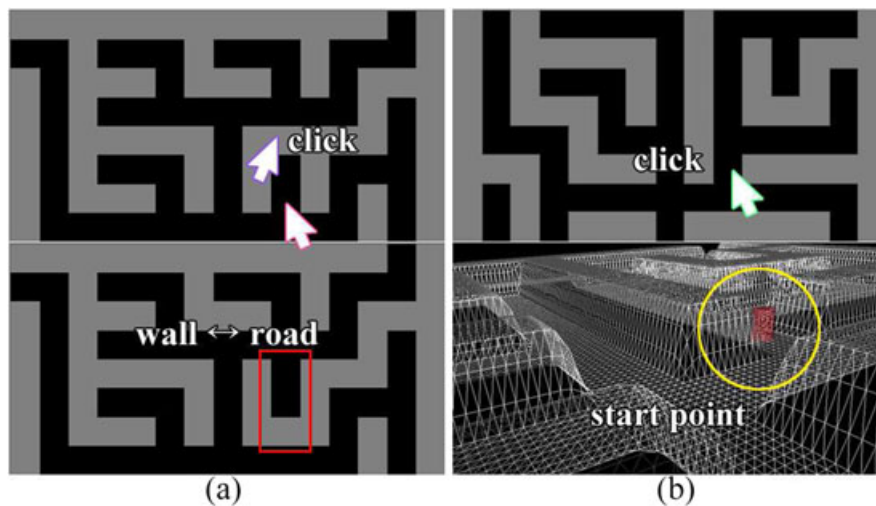


FIGURE 3 Interactive editing function of the authoring system: (a) maze pattern editing process and (b) setting starting point on the selected location

Here, ω^0 and ω^1 refer to the width and height of the authoring system window. When the cell of calculated coordinate P is a wall, it is transformed to a road cell and vice versa. m is the size of the 3D maze terrain, and v is the newly generated 2^n -sized maze texture map resolution. Figure 3 shows the edited maze pattern image that has been generated and the results of generating the starting point at the selected location.

The last function is sketch-based maze retrieval. The maze pattern generated through the proposed authoring

system has so far been limited to square pattern mazes. This research uses a sketching function to present an alternative and effectively handle diverse and complex maze patterns like circles and diagonal lines. In the proposed method, when the user draws a maze pattern directly on the grid, an image-processing algorithm is applied to transform the image into a maze texture map. This research defines the following simple rules for the generalized image processing of drawings:

1. The widths of the road and wall are the same.
2. Wall borderlines are drawn.
3. The inside of the borderlines is lightly colored.

Thus, the following image-processing procedures are applied to the drawn out maze. First, binarization is applied to differentiate the road and wall. As threshold values can differ, the authoring system UI provides the value. In the third defined rule, the results of binarization may not be accurate to the details of the coloring process. Therefore, a closing operation (morphology) is applied to fill the holes generated in

walls in the binarization process. The system allows the user to directly adjust the number of times the closing operation to be repeated. When a maze image is generated such that the sketch-based walls and roads are identified, it is transformed into a maze texture map. Figure 4 shows the processing of sketch-based maze retrieval.

The maze terrain authoring system is designed by integrating these three steps and can automatically generate various maze patterns upon request. It can also easily generate maze patterns desired by the user. The 3D terrain preview function of the authoring system is also available by coordinating the

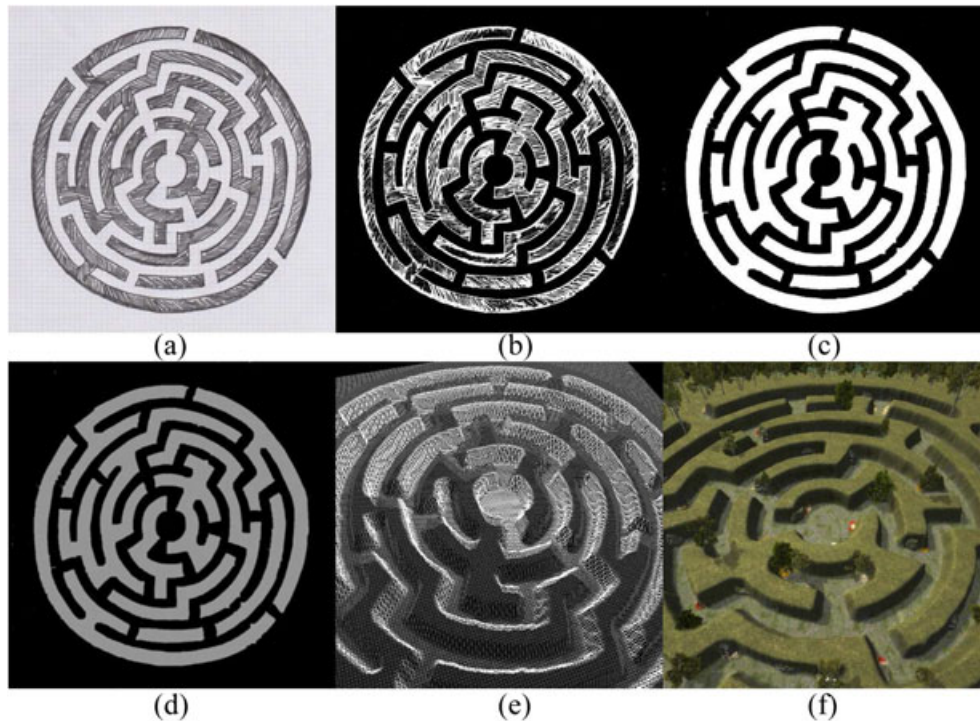


FIGURE 4 Generating maze texture maps using sketches: (a) sketch, (b) binarization, (c) closing operation (morphology), (d) texture map, (e) maze terrain (wireframe), and (f) complete scene

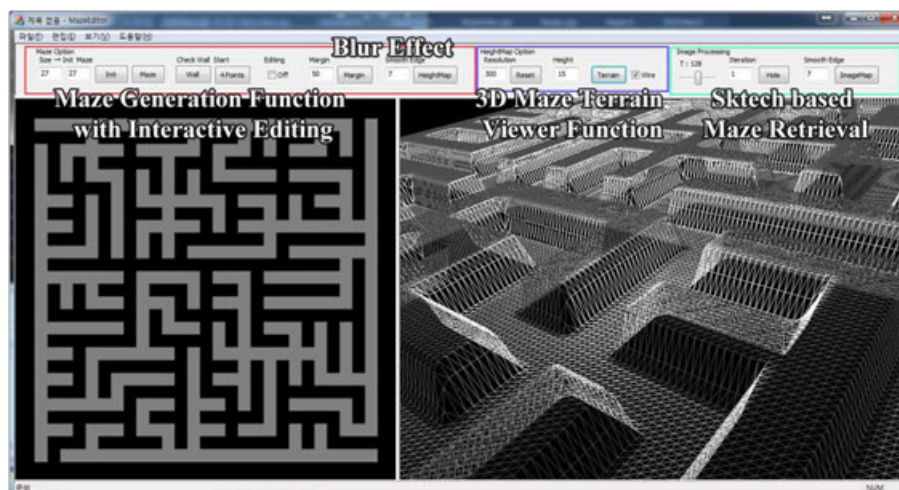


FIGURE 5 Structure of the proposed maze terrain authoring system

y-coordinate of plane vertex and the color value of the generated maze texture map. The blur is added to the maze texture map to enable smoothness adjustment to the generated terrain walls (Figure 5).

3.3 | Immersive interaction

The final aspect of MAVE is an immersive interaction that is suited for virtual maze environments. Here, the core objective is to materialize a portable system with simple structure that enables users to feel as if they are experiencing a real maze within a limited space. This study proposes an Arduino-based portable walking simulator that reacts to motions by detecting users walking in place.

The hardware for the proposed simulator is as follows. To design a simple and easy-to-wear portable simulator, core sensors are connected to the Arduino nano main board. An easy-to-wear ankle band is then developed. The motions of each leg are detected, and the Arduino structure is divided into master and slave to combine them into a single motion. In short, a band with the slave board is attached to both legs, and the user walks in place. Then, the master board receives information sent from the slave to determine the movement. The master then transmits the calculated motion to the virtual environment to reflect the user's motions. To transmit

data between master and slave boards, a Bluetooth module (HC-06) is used, and the gyro sensor attached to each slave board detects leg motions. As the virtual maze environment in this study establishes an integrated environment within a Unity3D engine, the calculated motion information is sent from the master to Unity3D via a serial port. Figure 6 shows the structure and data flow of the portable walking simulator.

The simulator slave board plays a crucial role in detecting the motion of walking in place. For detection, an MPU6050 gyro sensor, which measures six axes (X, Y, Z axis gradient + acceleration) was used in this research. This sensor is attached to the ankle band and detects yaw, pitch, and roll gradient; Figure 7 exhibits this process. When the user walks in place, the biggest gradient difference on the roll axis of the simulator is detected. When the user changes the views, the gradient of the pitch axis changes. Therefore, the simulator identifies walking in place through the gradient detected by the gyro sensor on both legs, which is then applied to the virtual environment.

Algorithm 2 is a process of recognizing a user walking in place of the proposed walking simulator. First, a change in the user's eye view is detected through the gradient of the pitch axis. If no change is detected, walking in place is recognized by calculating the gradient of the roll axis. The degree of change (T_{move}) in gradient of the roll axis of the leg may differ by user. Therefore, when the user wears the simulator



FIGURE 6 Structure of proposed portable walking simulator and data flow. HMD = head-mounted displays; MAVE = maze-based immersive virtual environment; SDK = software development kit

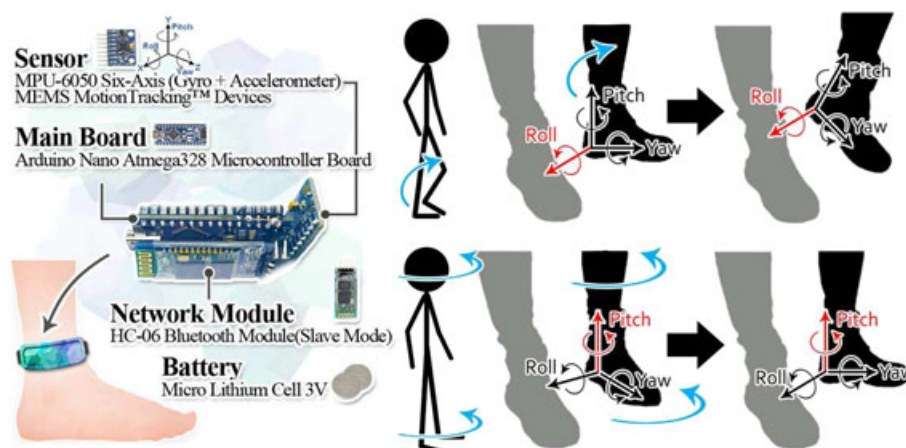


FIGURE 7 Detecting walking in place with the proposed simulator

and walks in place for about 10 s, the acceleration change is measured, and the average value of 20% minimum change is determined as the threshold value.

Algorithm 2 Identifying walking in place algorithm.

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1: state  $\leftarrow$  user's motion state.
2: ypr[3]  $\leftarrow$  yaw, pitch, roll gradient values of gyro sensor.
3:  $T_{rot}$   $\leftarrow$  threshold for identifying view changes (the pitch axis).
4:  $T_{move}$   $\leftarrow$  threshold for identifying movement (the roll axis).
5: procedure MOTION DETECTION state, ypr,  $T_{rot}$ ,  $T_{move}$ 
6: curr_ypr[3]  $\leftarrow$  current 3-axis gradient value.
7: pre_ypr[3]  $\leftarrow$  previous 3-axis gradient value.
8: curr_ypr[3] = ypr[3]
9: if abs(pre_ypr[1] - curr_ypr[1]) >  $T_{rot}$  then
10: state = 1 (change in eye view state).
11: else if abs(pre_ypr[2] - curr_ypr[2]) >  $T_{move}$  then
12: state = 2 (walking-in-place motion state).
13: else
14: state = 0 (pause state).
15: end if
16: pre_ypr = curr_ypr
17: end procedure
  
```

4 | EXPERIMENTAL RESULTS AND ANALYSIS

The proposed MAVE authoring system used Microsoft Visual Studio 2013, DirectX SDK 9.0c (Microsoft, Redmond, WA, USA), and OpenCV 2.4.11; our walking simulator used Arduino sketch v.1.6.12 (Interaction Design Institute Ivrea (IDII), Ivrea, Italy). Unity3D 5.3.4f1 (Unity Technologies, San Francisco, CA, USA) and Oculus SDK (ovr_unity_utilities 1.3.2, Oculus, Irvine, CA, USA) were used to establish the MAVE integrated environment. The PC used in the MAVE tests has the following specifications: Intel Core i7-6700 CPU (Intel Corporation, Santa Clara, CA,

USA), 16GB RAM, Geforce GTX 1080 GPU (NVIDIA, Santa Clara, CA, USA).

This study confirms whether the MAVE authoring system effectively generates virtual maze environment. This process is shown in Figure 8. Using its maze-generating algorithm and interactive editing, the authoring system enables easy and quick calculation of finite maze patterns in intuitive structures and automatically creates them into 3D maze terrain through maze texture map transformation. Adding graphic factors that fit the overall concept can generate various virtual maze environments. Moreover, sketch-based maze retrieval can also effectively generate unique maze patterns for creating maze terrains.

The technical performance of MAVE is analyzed; for MAVE's virtual scenes to be used as VR content without VR sickness, it is recommended to have no more than 75 frames per second and the number of total polygons of the rendered scene to not exceed 2M. Otherwise, a user wearing an HMD can feel dizzy just by looking at the rendered virtual scene. Table 1 shows the average fps and polygon number calculated 10 times over the three scenes in Figure 8 from the starting point to destination. Whereas many polygons (3.900M) were recorded in certain camera angles, it can be confirmed that the technical specifications of the generated MAVE scenes meet PC platform VR standards.

Detailed tests on MAVE were carried out through questionnaires. The survey asked general users whether the MAVE virtual scene provides a novel sense of presence and experience as well as VR sickness. The survey was given to 20 participants between the ages of 21 and 35 in a MAVE experience space (Figure 9).

The first survey consisted of three subjective questions for analyzing overall satisfaction and immersion (Table 2). Questionnaires developed to compare presence between reality and VR include the Witmer and Singer Presence questionnaire and that developed by Slater, Usoh, and Steed.²⁰ These revealed significant results when reality and VR are compared. However, comparison between different VR systems

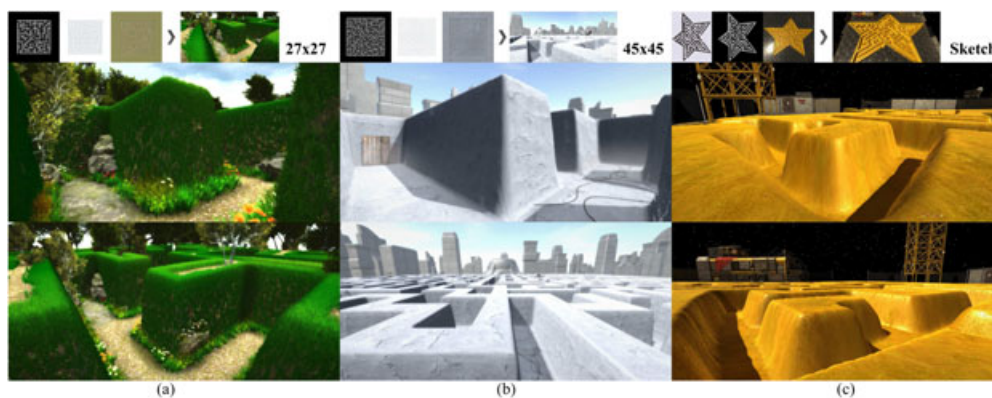


FIGURE 8 Maze-based immersive virtual environment virtual scene results generated via the proposed authoring system: (a) forest, 27×27 , (b) deserted city, 45×45 , and (c) space, sketch

TABLE 1 Analysis results of specifications of maze-based immersive virtual environment virtual scene

	Frames per second (fps)			Number of polygons (M)		
	Minimum	Maximum	Mean	Minimum	Maximum	Mean
Forest	77.6	92.9	88.3	0.104	3.900	1.183
Deserted City	97.4	120.7	100.8	0.275	1.700	0.855
Space	86.5	88.1	87.4	0.127	0.633	0.395

**FIGURE 9** Maze-based immersive virtual environment experience**TABLE 2** MAVE satisfaction and immersion analysis results

Question Phase	Standard	Minimum	Maximum	Mean (SD)
(a) New presence and experience provided by MAVE scene				
Forest	3(Norm)	2	5	4.055(0.79)
Deserted city	3(Norm)	3	5	4.020(0.65)
Space	3(Norm)	2.5	5	4.125(0.74)
(b) Realism that feels like walking in virtual environment				
Only K	5(Real)	2	4.5	3.27(0.91)
Only S	5(Real)	2.5	5	3.67(0.83)
K → S	5:5(K:S)	1:3	3:5	2.35 : 3.95
S → K	5:5(K:S)	1:3	4:5	2.43 : 4.15
Wilcoxon result (<i>P</i> value)			8.37822e-05	
(c) Motion accuracy through walking in place in reality and virtual space				
Deserted city	3 (Accu-rate)	1.5 (Slow)	4 (Fast)	2.665(.578)

Note. K = keyboard; MAVE = maze-based immersive virtual environment; S = simulator; SD = standard deviation.

is meaningless. Therefore, this study conducted a subjective survey to confirm presence and immersion and performed a statistical experiment to verify the objectivity of the experimental results. To the question of whether MAVE's three virtual spaces provide presence and experience, most respondents answered with a high score above 4.0. The second

question compared and analyzed the relationship between interaction and immersion either a general PC input device (keyboard, K) and the proposed simulator (S). The interaction experience groups were divided into four (only K, only S, K after K, K after S) and were analyzed in detail. The simulator was found to provide users with a similar feeling of

TABLE 3 SSQ analysis results of proposed simulator

		Mean/(16 items)	SD	Minimum	Maximum	Rate
Total	K	6.35(0.40)	6.19	0	22	10%
	S	5.7(0.36)	5.17	0	19	10%
Nausea	K	1.4(0.09)	1.56	0	4	0%
	S	1.4(0.09)	1.62	0	7	5%
Oculomotor	K	3.5(0.22)	3.14	0	10	20%
	S	3.25(0.20)	2.32	0	8	15%
Disorientation	K	3.5(0.22)	3.75	0	12	20%
	S	3.25(0.2)	3.31	0	12	20%
After 180 sec	K	7.2(0.45)	7.5	0	26	20%
	S	5.85(0.37)	4.9	1	20	10%

Note. K = keyboard; SD = standard deviation; S = simulator; SSQ = simulator sickness questionnaire.

walking even in a virtual maze environment. In fact, comparison between K and S through Wilcoxon test showed the significance probability at $8.37822e-05$, which is smaller than significance level (.05), indicating that the null hypothesis can be rejected. As a result, it was statistically proven that the proposed simulator shows better realism. When both input devices were used, the difference in realism was even more clear. Lastly, motion accuracy was tested as the user walked in place. When the user felt a timing difference between the real and virtual, immersion is reduced and VR sickness can occur. The test results, 2.662 (0.578), show that it was slightly slow in reaction, though it was not inconvenient to the point that motion in the virtual environment was affected. Therefore, the study confirmed that relatively appropriate movement was being reflected.

Second, VR sickness caused by MAVE was analyzed systematically. This study utilized the simulator sickness questionnaire (SSQ),²¹ which analyzes of various symptoms that users can experience within VR systems through diverse tests. Sixteen items that can systematically measure simulator sickness were selected. When the respondent inputs one of the four possible answers (none, slight, moderate, and severe) for 16 items, the total score calculates the objective performance of the simulator. SSQ can analyze in detail areas including nausea (items: 1, 6–9, 15, and 16); oculomotor symptoms (items: 1–5, 9, and 11); and disorientation (items: 5, 8, 10, 11, and 12–14). Table 3 shows the result of SSQ tests for the proposed simulator. As this research requires objective analysis of the MAVE virtual scene and simulator, the results were analyzed through raw data excluding weight.²² VR sickness caused by MAVE virtual scenes was divided into keyboard (K) and simulator (S) and was measured twice with two different questionnaires. Half of the participants experienced the simulator first whereas the other half experienced the simulator last. With overall average scores of 6.35 and 5.7 (0.4 and 0.36, below slight), the MAVE virtual scenes and simulator had a very slight impact on VR sickness. A small ratio

of participants gave over 16 points, recording 10% for both categories. Detailed factor analysis showed a relatively higher score for oculomotor symptoms and disorientation, which can be translated as the effect of a sense of repetition and direction resulting from the maze terrain. One important point here is that the simulator gave a sense of realism that made users feel as if they were walking in a maze terrain, thereby reducing VR sickness. The gap was found to be greater for participants who experienced the simulator first, probably because sickness was maximized when using the keyboard after the experience owing to the fast scene changes without a sense of realism. Lastly, it was also found that after staying in the maze terrain for a longer time (180 s), the simulator had a more positive effect for VR sickness (7.2, 5.85).

5 | LIMITATION

MAVE consists of two structures: an authoring system for maze scenes and a portable walking simulator. Each system has limitations. First, the authoring system can only automatically generate square finite maze patterns. The alternative method proposed for overcoming this limitation, sketch-based maze retrieval, does not provide a UI for hand drawing in the authoring system. Moreover, MAVE scenes with graphic factors cannot be generated in the authoring system. The simulators are aimed at providing enhanced immersion compared to a keyboard interface and use a simple structure rather than a complex motion platform. However, the inconvenience caused by marching in place and motion diversity, like walking backwards, needs to be improved in future research.

6 | CONCLUSION AND FUTURE WORK

This study proposes MAVE, an immersive virtual environment that provides users with a sense of presence and

experience. MAVE consists of a maze terrain authoring system that can effectively generate various maze patterns in a user-oriented structure, and a portable walking simulator that is suitable for enhancing interactive immersion in a virtual maze environment. Through the three-step maze-generation algorithm, interactive editing, and sketch-based maze retrieval, the authoring system can automatically generate new maze patterns desired by the user intuitively and quickly. The generated 2D maze pattern is then transformed to effectively reconstruct a 3D maze terrain. MAVE's portable walking simulator was designed for interactivity, with enhanced immersion. Using an Arduino-based gyro sensor, the simulator detects marching in place to control the user's motion within the virtual maze environment. This study also confirmed that MAVE can generate a virtual scene that provides users with a novel sense of presence and experience, while generating a virtual environment that enhances immersion. Tests ranging from technical performance analysis demanded by the VR system to surveys targeting general users were carried. Questionnaires helped confirm that MAVE virtual maze scenes and simulator-based interaction reduce VR sickness while increasing user satisfaction.

Future research will include more functions for the authoring system, so that generating all MAVE scenes within the system could be plausible through an integrated system. As for maze generation, the limitations of the current authoring system will be overcome to automatically generate various maze terrains and expand user interactivity. Lastly, as for the walking simulator, a comparison test with a treadmill motion platform will be helpful in applying the system to various VR content.

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